Artifact Appendix: PrivAudit

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A Artifact Appendix

A.1 Abstract

This artifact is the code and data for PrivAudit, a dual-lens audit of website privacy practices under the CCPA. The first lens scores each privacy policy with an LLM against a CCPA-based rubric. The second records the cookies a site sets under six browser privacy configurations. The artifact joins the two per website. The datasets and the CCPA-subjectivity labels are inside `data/data.zip`, so every result in the paper is reproduced from the bundle alone, with no scraping or crawling. Each result has one script that prints its numbers, and `README.md` maps each claim to its script. Everything runs on a laptop in seconds. The artifact is licensed CC BY 4.0.

A.2 Description & Requirements

The repository root holds `README.md`, `requirements.txt`, and `data/` (the data bundle). `analysis/` has the scripts for the paper's tables and figures (`analysis/figures/` for the plots). `policy_analysis/` scores policies with the LLM. `crawler/` runs the cookie measurements. `tranco_scale/`, `inner_page/`, `subjectivity/`, `banner_detection/`, and `cookie_categorization/` hold the other studies.

A.2.1 Security, privacy, and ethical concerns. Reproducing the results only reads the bundled data. It does not contact any website, and it needs no special privileges. The data contain cookie names and values from public homepages and public privacy policies, and no user data. The optional crawler visits each site once per configuration, in round-robin order.

A.2.2 How to access. Zenodo (permanent, CC BY 4.0): concept DOI `10.5281/zenodo.22681470`, which always resolves to the latest version. The version evaluated by the committee is `10.5281/zenodo.22682191` (tag `ccs-ae-v2`). A later version, `10.5281/zenodo.23674179` (tag `ccs-ae-v3`), has two script fixes made after the evaluation: the CCPA-mention row of Table 1 and the group labels in the split-violin figure. Source: `https://github.com/r-andlab/PrivAudit`.

A.2.3 Hardware dependencies. None. An ordinary laptop; the unpacked data take about 200 MB.

A.2.4 Software dependencies. Python 3.10+ (tested with 3.13) with pandas, numpy, scipy, statsmodels, matplotlib, and seaborn (`requirements.txt`). No internet is needed. Node.js 18+, Chrome, and an OpenAI API key are needed only to re-run the crawler and the LLM scoring, which is optional.

A.2.5 Benchmarks. `data/data.zip` (26 MB) holds the cookie measurements for the six configurations, the extracted policy claims, the re-verified CCPA-subjectivity labels, and the inner-page and Tranco crawl data. The corpus is 998 websites (602 CCPA-Subject, 396 Not-Subject).

A.3 Set Up

A.3.1 Installation.

```
git clone https://github.com/r-andlab/PrivAudit
cd PrivAudit && git checkout ccs-ae-v3
pip install -r requirements.txt
cd data && unzip data.zip && cd ..
```

(Or unzip the Zenodo archive.) Scripts find `data/` relative to the repository root; set `PRIVAUDIT_ROOT` to use another path.

A.3.2 Basic test. Run `python3 analysis/apply_verified_labels.py` once before the experiments. It should print `cookie sites 919 -> 915 | Subject 552 / Not-Subject 363`. After that, every script reads only from `data/`.

A.4 Evaluation Workflow

A.4.1 Major claims.

- (C1): CCPA-Subject sites give stronger disclosures than Not-Subject sites (opt-out 77% vs. 57%, access 85% vs. 68%, delete 86% vs. 69%, GPC 29% vs. 13%). Shown by (E1); §4.1, Table 1.
- (C2): Cookie tracking is pervasive in both groups: 6,392 Targeting cookies, 49% third-party writes. Shown by (E2); §4.2, Table 2.
- (C3): Privacy signals reduce but do not stop tracking: GPC cuts Targeting cookies by 55.8% but third-party Targeting by about 41%; uBlock cuts 87%. Shown by (E2); §4.2.
- (C4): Linking the lenses exposes gaps: of 110 GPC-claiming Subject sites that set Targeting cookies, 39.1% show no reduction; 50.8% of sites that disclose they do not sell data still track; 9.7% of 269 third-party-Targeting setters disclose no data collection. Shown by (E3); §4.3.
- (C5): The results hold up: 14 of 25 tests survive Holm correction, the gap persists after controlling for firm size, and the pattern repeats on a 1,000-site Tranco corpus. Shown by (E4)–(E6); §4.1, §4.4.

A.4.2 Experiments. Each experiment is under 5 person-minutes and under 5 compute-seconds, on a laptop. Run all from the repository root, after the basic test.

- (E1): [C1, Table 1] `python3 analysis/regen_table1_policy.py`. Expect (Subject/Not-Subject): opt-out 77.1/57.1, access 85.4/68.3, delete 85.9/69.4, GPC 28.7/13.1 (N=567/366). The

"CCPA" mention row is computed from the shipped policy text and is approximate: about 66/43 against the paper's 67.3/43.1, over the 551/350 policies whose text is shipped.

(E2): [C2, C3] `python3 analysis/regen_overall_after_exclusion.py` and `regen_table_3p_reduction.py`. Expect 19,119 cookies on 913 sites, 6,392 Targeting, 58.5%/41.5% first/third-party, GPC Targeting reduction 55.8%, uBlock 87.2%.

(E3): [C4] `python3 analysis/regen_joint_lens.py` and `compliance_matrix_1798130.py`. Expect 163 GPC claims, 110 set Targeting cookies in Default; 58 (52.7%) reduce under 20%, 43 (39.1%) not at all; 204 do-not-sell sites, 95 (50.8%) track; 269 setters, 26 (9.7%) and 22 (8.2%) silent.

(E4): [C5] `python3 analysis/regen_mht_table.py`. Expect Holm-significant 14/25.

(E5): [C5] `python3 analysis/size_control_analysis.py`. Expect large firms (N=563): opt-out 77% vs. 46%, access and delete 85% vs. 60%.

(E6): [C5] `python3 tranco_scale/analyze_ccpa_frame.py`. Expect 1,000 sites, 72% usable policy, 28% with an opt-out signal, 54.2% of those still set a third-party Targeting cookie.

(E7): [Figures] `python3 analysis/figures/plot_split_violin_updated.py`, `plot_cdf_comprehensive_banner_ccpa.py`, `plot_ecdf_lifespan_ccpa_split.py`. Expect PDF and PNG files in `figures_output/` matching the paper's Figures (banner CDF: N=183 Subject, 56 Not-Subject).

A.5 Notes on Reusability

PrivAudit can audit a new list of websites. The two lenses are separate: `policy_analysis/` scrapes and scores policies, and `crawler/` records cookies under the six configurations (put target domains in a CSV with an `AccessibleDomain` column and point a config in `crawler/configs/` at it). Joining the outputs with `regen_joint_lens.py` gives the audit signals. The LLM step needs an OpenAI API key; the README has details.

A.6 Version

Based on the LaTeX template for Artifact Evaluation V20220926.